THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS CHAPTER TEST SET B

Chapter 14: **Area** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The area of a rectangle of length 15 cm and breadth 8 cm is: 1

(a) 120 cm² **(b)** 46 cm² **(c)** 23 cm² **(d)** 128 cm²

2. The area of a triangle with base 14 cm and height 9 cm is: 1

(a) 63 cm² **(b)** 77 cm² **(c)** 126 cm² **(d)** 23 cm²

3. The area of a parallelogram with base 12 cm and height 5 cm is: 1

(a) 60 cm² **(b)** 34 cm² **(c)** 65 cm² **(d)** 30 cm²

4. The area of a trapezium with parallel sides 10 cm and 14 cm and height 6 cm is: 1

(a) 72 cm² **(b)** 140 cm² **(c)** 144 cm² **(d)** 82 cm²

5. The area of a trapezium with parallel sides 12 cm and 18 cm and height 9 cm is: 1

(a) 270 cm² **(b)** 216 cm² **(c)** 147 cm² **(d)** 135 cm²

6. The area of a rhombus with diagonals 18 cm and 14 cm is: 1

(a) 252 cm² **(b)** 32 cm² **(c)** 126 cm² **(d)** 63 cm²

7. A rectangle has area 200 cm² and length 25 cm. Its breadth is: 1

(a) 7 cm **(b)** 8 cm **(c)** 175 cm **(d)** 10 cm

8. A diagonal of a rectangle divides it into two triangles of: 1

(a) equal area **(b)** double area **(c)** unequal area **(d)** no area

9. If the base of a triangle is doubled and height kept the same, the area: 1

(a) stays the same **(b)** becomes four times

(c) halves **(d)** doubles

10. The height of a parallelogram of area 72 cm² and base 12 cm is: 1

(a) 24 cm **(b)** 60 cm **(c)** 12 cm **(d)** 6 cm

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Find the area of a trapezium with parallel sides 9 cm, 13 cm and height 7 cm. 2

12. A rhombus has diagonals 12 cm and 16 cm. Find its area. 2

13. Find the base of a parallelogram of area 108 cm² and height 9 cm. 2

14. How many 25 cm × 20 cm tiles cover a floor of 5 m by 4 m? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. Find the area of a trapezium with parallel sides 12 cm and 20 cm and distance between them 8 cm. Also find the area of an equal-height rectangle with width 20 cm. 3

16. Find the area of a trapezium with parallel sides 9 cm and 15 cm and distance between them 10 cm. Also find the area of an equal-height rectangle with width 15 cm. 3

17. Find the area of a trapezium with parallel sides 14 cm and 22 cm and distance between them 7 cm. Also find the area of an equal-height rectangle with width 22 cm. 3

OR

Find the area of a trapezium with parallel sides 18 cm and 30 cm and distance between them 12 cm. Also find the area of an equal-height rectangle with width 30 cm. 3

18. Find the area of a trapezium with parallel sides 10 cm and 26 cm and distance between them 9 cm. Also find the area of an equal-height rectangle with width 26 cm. 3

OR

Find the area of a trapezium with parallel sides 16 cm and 24 cm and distance between them 15 cm. Also find the area of an equal-height rectangle with width 24 cm. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 80–100 words each.

19. **Case study:** A rhombus-shaped tile has diagonals 24 cm and 18 cm. 5

(i) Find the area.

(ii) Find the side.

(iii) How many tiles to cover 5.4 m² (ignore waste)?

OR

(i) A rectangle's length is increased by 20% and its breadth decreased by 20%. Find the percent change in area. 5

(ii) Explain.

20. (i) Find the area of a quadrilateral with a diagonal of 20 cm and offsets 7 cm and 9 cm to the diagonal. 5

(ii) Name the method.

OR

Case study: A rectangular hall 12 m × 8 m has a carpet in the middle leaving a 1 m border. 5

(i) Area of the carpet.

(ii) Area of the border.

(iii) Cost of polishing the border at ₹ 55/m².

END OF QUESTION PAPER — Total: 40 Marks